

Partial Heart Transplant in a Neonate With Irreparable Truncal Valve Dysfunction

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IMPORTANCE The treatment of neonates with irreparable heart valve dysfunction remains an unsolved problem because there are no heart valve implants that grow. Therefore, neonates with heart valve implants are committed to recurrent implant exchanges until an adult-sized valve can fit.

OBJECTIVE To deliver the first heart valve implant that grows.

DESIGN, SETTING, AND PARTICIPANTS Case report from a pediatric referral center, with follow-up for more than 1 year. Participants were a recipient neonate with persistent truncus arteriosus and irreparable truncal valve dysfunction and a donor neonate with hypoxic-ischemic brain injury.

INTERVENTION First-in-human transplant of the part of the heart containing the aortic and pulmonary valves.

MAIN OUTCOMES AND MEASURES Transplanted valve growth and hemodynamic function.

RESULTS Echocardiography demonstrated adaptive growth and excellent hemodynamic function of the partial heart transplant valves.

CONCLUSIONS AND RELEVANCE In this child, partial heart transplant delivered growing heart valve implants with a good outcome at age 1 year. Partial heart transplants may improve the treatment of neonates with irreparable heart valve dysfunction.

JAMA. 2024;331(1):60-64. doi:10.1001/jama.2023.23823

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The treatment of neonates with irreparable heart valve dysfunction remains an unsolved problem because there are no heart valve implants that grow.¹ The standard of care for neonatal heart valve implants are cadaver homografts. However, cadaver homografts are not viable and therefore lack the capacity for growth or self-repair. This commits neonates with cadaver homografts to recurrent implant exchanges until an adult-sized valve can fit. As a result, clinical outcomes are poor. For example, neonatal truncal valve implants have more than 50% mortality in infancy.² In contrast, neonatal heart transplants have only 15% mortality in infancy³ because transplanted hearts grow.^{4,5} However, long-term outcomes of neonatal heart transplants are limited by inevitable ventricular dysfunction, with 50% mortality by 20 years.³

Partial heart transplant is a new approach to deliver growing neonatal heart valve implants that involves transplant of the part of the heart containing the outflow valves only, while the native ventricles are spared.⁶ Partial heart transplants differ from cadaver homografts because they are treated like heart transplants to maintain viability of the valves. This allows partial heart transplants to grow just like the valves in heart transplants (Table). Here we report the first partial heart transplant with neonatal donor aortic and pulmonary valves to treat a neonate with persistent truncus arteriosus and irreparable truncal valve dysfunction.

Table. Differences Between Partial Heart Transplants, Homografts, and Heart Transplants

	Partial heart transplant	Homograft	Heart transplant
Donor	Organ donor	Cadaver	Organ donor
Graft	Heart valve	Heart valve	Whole heart
ABO typing	Yes	No	Yes
Preservation	Hours	Years	Hours
Immunosuppression	Yes	No	Yes
Grows with child	Yes	No	Yes

Methods

Recipient

A male neonate with prenatally diagnosed persistent truncus arteriosus and severe truncal valve dysfunction was delivered at gestational age 38 weeks 2 days. Birth weight measured 2.6 kg. Postnatal echocardiography confirmed type A2 persistent truncus arteriosus (Figure 1A). The truncal valve annulus measured 10 mm, and the quadricuspid leaflets were thickened and dysplastic. There was severe truncal valve insufficiency (Figure 2), with flow reversal in the transverse aortic arch. The outlet ventricular septal defect had basal muscular extension.

Biventricular function was initially normal. The recipient's blood type was O.

The recipient's preoperative course was notable for progressive ventricular dysfunction, requiring escalating inotropic support. Trophic feeds had to be withheld due to concern for necrotizing enterocolitis. All possible treatment options were reviewed in multidisciplinary conferences before a shared decision was made with the parents to pursue partial heart transplant. High-fidelity 3D printed models were used for surgical planning.⁷ On day-of-life 18 of the recipient, a suitable neonatal donor was identified in another state.

Donor

The donor's parents were consented by the local organ procurement organization. The donor was a 2-day-old female weighing 3.7 kg. Delivery had been complicated by hypoxic-

Key Points

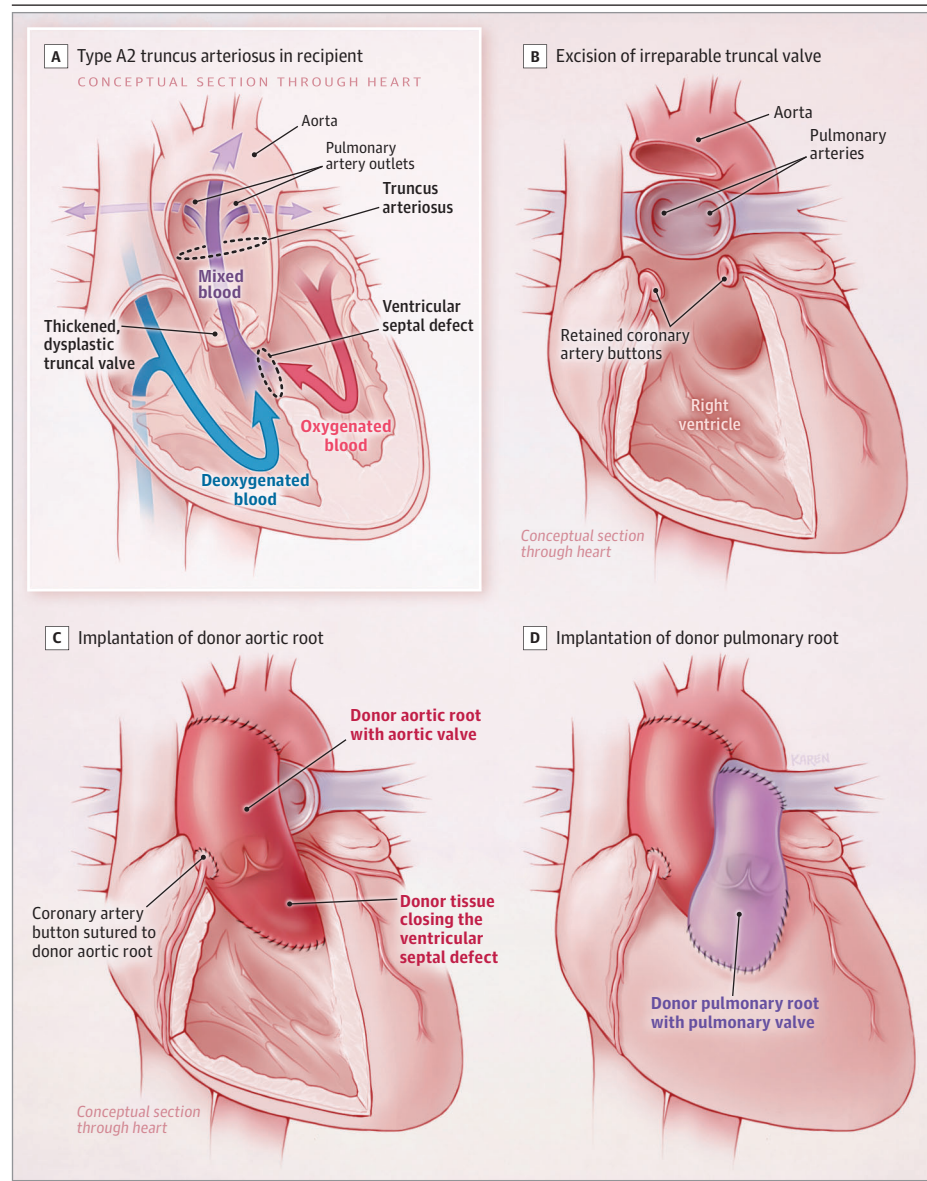
Question Do partial heart transplants deliver growing heart valve implants for neonates?

Findings The first-in-human partial heart transplant demonstrated valve growth during follow-up for more than 1 year.

Meaning Partial heart transplant is a new approach to treat neonates with irreparable heart valve dysfunction.

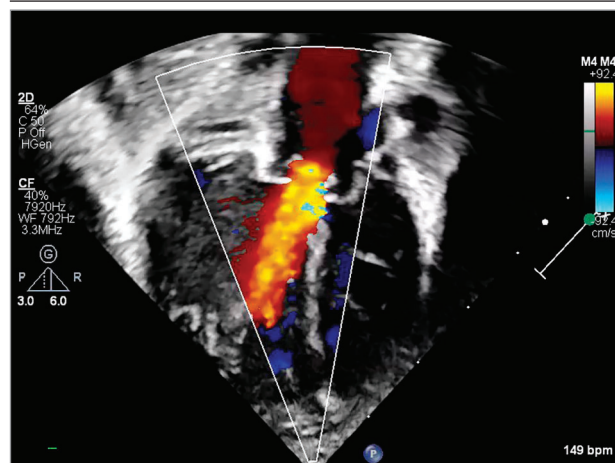
ischemic brain injury. The donor's blood type was B. Echocardiography showed structurally normal outflow valves with excellent function. The aortic valve annulus measured 7 mm and the pulmonary valve annulus measured 8 mm. The heart was

Figure 1. Partial Heart Transplant



Recipient's anatomy with truncus arteriosus and a dysplastic, irreparable truncal valve (panel A), excision of the irreparable truncal valve (panel B), implantation of the donor aortic root with closure of the ventricular septal defect using donor tissue (panel C), and implantation of the donor pulmonary root (panel D).

Figure 2. Truncal Valve Regurgitation



The echocardiogram shows the dysplastic truncal valve with severe truncal valve regurgitation (orange jet).

donated after cardiac death. Following expedient sternotomy, the heart was procured using standard surgical techniques. On the back table, the pulmonary root was dissected in the fashion of a pulmonary autograft. The aortic root was dissected with donor tissue for closure of the recipient ventricular septal defect.

Statistics

Spearman ρ was used to test the hypothesis that age and valve diameter had a positive monotonic relationship (Prism 10; GraphPad Software Inc).

Results

Partial Heart Transplant Operation

Informed consent was obtained from the recipient's parents. The recipient operation (Video 1) involved sternotomy, cardiopulmonary bypass, and cardioplegic arrest of the heart. The arterial trunk was transected, and irreparable truncal valve dysfunction was confirmed by direct examination of the valve. The pulmonary artery ostia and coronary artery buttons were dissected, and the irreparable truncal valve was excised (Figure 1B). The donor aortic root was transplanted first, using donor tissue to close the ventricular septal defect. The coronary artery buttons were reimplanted (Figure 1C). The right ventricular outflow tract was enlarged, and the pulmonary root was transplanted (Figure 1D). Ischemic time was 395 minutes; cardiopulmonary bypass time, 197 minutes; aortic cross-clamp time, 135 minutes; and total operation time, 389 minutes.

Postoperative Care

Immunosuppression of the recipient followed a modified protocol for ABO-incompatible pediatric heart transplant. Briefly, mycophenolate mofetil (15 mg/kg) and solumedrol (10 mg/kg) were given preoperatively. Solumedrol (10 mg/kg)

was repeated on release of the cross-clamp. Postoperatively, antithymocyte globulin (1.5 mg/kg) was given once for ABO incompatibility. Solumedrol (5 mg/kg every 8 hours) was given for 6 doses and the dose was then tapered over 2 weeks. Mycophenolate mofetil (15 mg/kg) and tacrolimus titrated to trough levels of 5 to 8 ng/mL were given every 12 hours. Rejection was monitored using donor-derived cell-free DNA (AlloSure Heart; CareDx), which remained below 0.15% 12 weeks postoperatively. Anti-B titers remained low.

The early postoperative course was notable for a hyperdynamic circulation treated with milrinone and nicardipine. The patient was extubated on postoperative day 6. All hemodynamic drips were weaned by postoperative day 17. Feeding intolerance was treated with a gastrostomy tube. The patient was discharged on postoperative day 30.

Follow-up

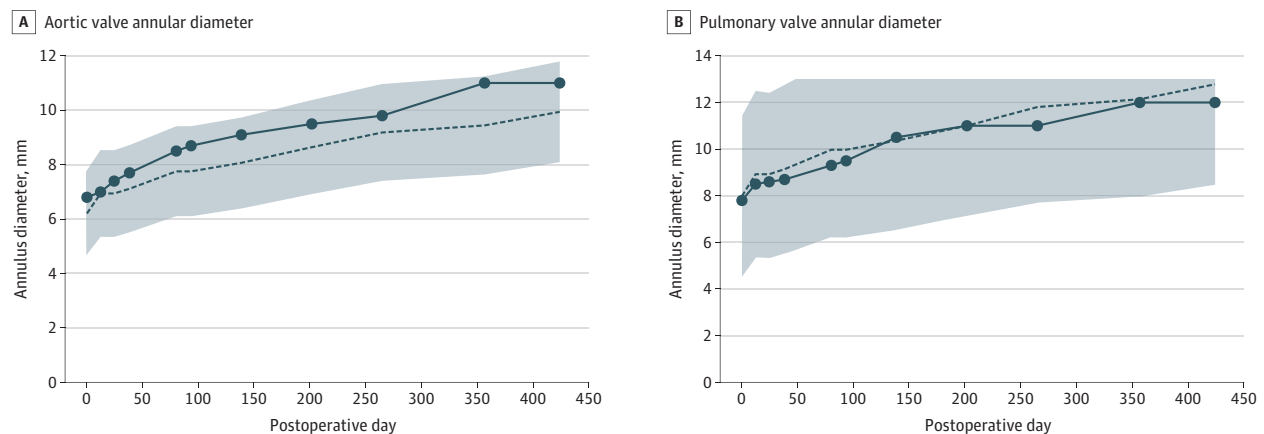
Serial echocardiograms have shown adaptive growth of the transplanted aortic valve ($\rho = 0.97$, $P < .001$) and pulmonary valve ($\rho = 0.96$, $P < .001$) (Figure 3). On follow-up at age 14 months, echocardiography showed no obstruction and no insufficiency of the transplanted aortic and pulmonary valves (Video 2). The patient's developmental milestones included playing, looking for objects, crawling, standing, and making different sounds but not babbling. He received tube feeds for oral aversion. At examination, height was 71 cm and weight 8.2 kg.

Discussion

The status quo for approaches to deliver growing neonatal heart valve implants is based on tissue engineering.⁸ However, tissue engineering has failed in clinical translation for decades. We developed a new approach to deliver growing neonatal heart valve implants that is based on transplantation.⁶ The rationale for partial heart transplant is that pediatric heart transplants grow.^{4,5} Moreover, failure of heart transplant outflow valves is exceedingly rare.⁹⁻¹¹ While heart transplant long-term outcomes are limited by inevitable ventricular dysfunction,¹² partial heart transplants spare the native ventricles and are therefore expected to last a lifetime.

These advantages need to be balanced against the disadvantages inherent in transplantation, namely limited donor availability and risks from immunosuppression. Donor availability for partial heart transplants is substantially higher than for heart transplants because grafts from neonatal donors with ventricular dysfunction and domino grafts from heart transplant recipients with functional valves can be used.¹³ Immunosuppression for partial heart transplants is based on immunosuppression for pediatric heart transplants, because the valves contained in pediatric heart transplants are known to grow.^{4,5} It can be less stringent because the valves in heart transplants are spared from rejection.^{9,10,14} The long-term risks from immunosuppression are predictable based on the experience with pediatric heart transplants. In infant heart transplant recipients, the

Figure 3. Partial Heart Transplant Growth



The black circles show postoperative aortic and pulmonary valve annular diameters measured by echocardiography. The dotted lines show z score 0, which corresponds to the mean valve annular diameters of healthy infants with

the same body surface area. The shaded areas show z scores -2 to $+2$, which include 95% of valve annular diameters in healthy infants with the same body surface area.

major long-term risks from immunosuppression are post-transplant lymphoproliferative disease caused by Epstein-Barr virus infection (12% by 10 years) and severe kidney dysfunction caused by calcineurin inhibitors (6% by 10 years).¹² Importantly, stopping immunosuppression for a partial heart transplant would simply turn the implant into a nongrowing homograft.¹⁵

Limitations

Given the complications observed with other treatments for irreparable heart valve dysfunction in neonates, longer-term

follow-up and a larger cohort of patients are needed to determine whether partial heart transplant confers advantages for long-term survival and reduced morbidity.

Conclusion

In this child, partial heart transplant delivered growing heart valve implants with a good outcome at age 1 year. Partial heart transplant may improve the treatment of neonates with irreparable heart valve dysfunction.

ARTICLE INFORMATION

Accepted for Publication: October 24, 2023.

Author Contributions: Drs Turek and Rajab had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Turek, Kang, Overbey, Rajab.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: All authors.

Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Turek, Rajab.

Administrative, technical, or material support: Turek, Kang, Overbey, Carboni, Rajab.

Supervision: Turek, Rajab.

Conflict of Interest Disclosures: Dr Rajab reported receiving National Institutes of Health (NIH)/National Heart, Lung, and Blood Institute grant HL169059 (a Small Business Technology Transfer grant, in collaboration with Tissue Testing Technologies LLC, for a method to prolong storage of partial heart transplant grafts). No other disclosures were reported.

Funding/Support: Preclinical research leading to this case report was supported by the Brett Boyer Foundation.

Role of the Funder/Sponsor: The Brett Boyer Foundation had no role in the design and conduct

of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See the [Supplement](#).

Additional Contributions: We gratefully acknowledge the contributions of the operative teams, which included 3 pediatric cardiac surgeons, 1 surgical fellow, 3 surgical residents, 1 medical student, 1 cardiology attending, 2 anesthesia attendings, 1 anesthesia resident, 5 nurses, and 2 cardiovascular perfusion specialists. Dr Turek led the recipient operation and perioperative care. Dr Kang assisted in the donor operation. Dr Overbey assisted in perioperative care. Dr Carboni read the echocardiograms and assisted in perioperative care. Dr Rajab led the donor operation. Mathew J. Gregoski, PhD (South Carolina Clinical & Translational Research Institute), provided statistical consultation. Dr Gregoski received no compensation for his contributions. We thank the patient's father for providing permission to publish the child's case.

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